

FormationCreator: Designing A VR Dance Formation System for Intangible Cultural Heritage Dance

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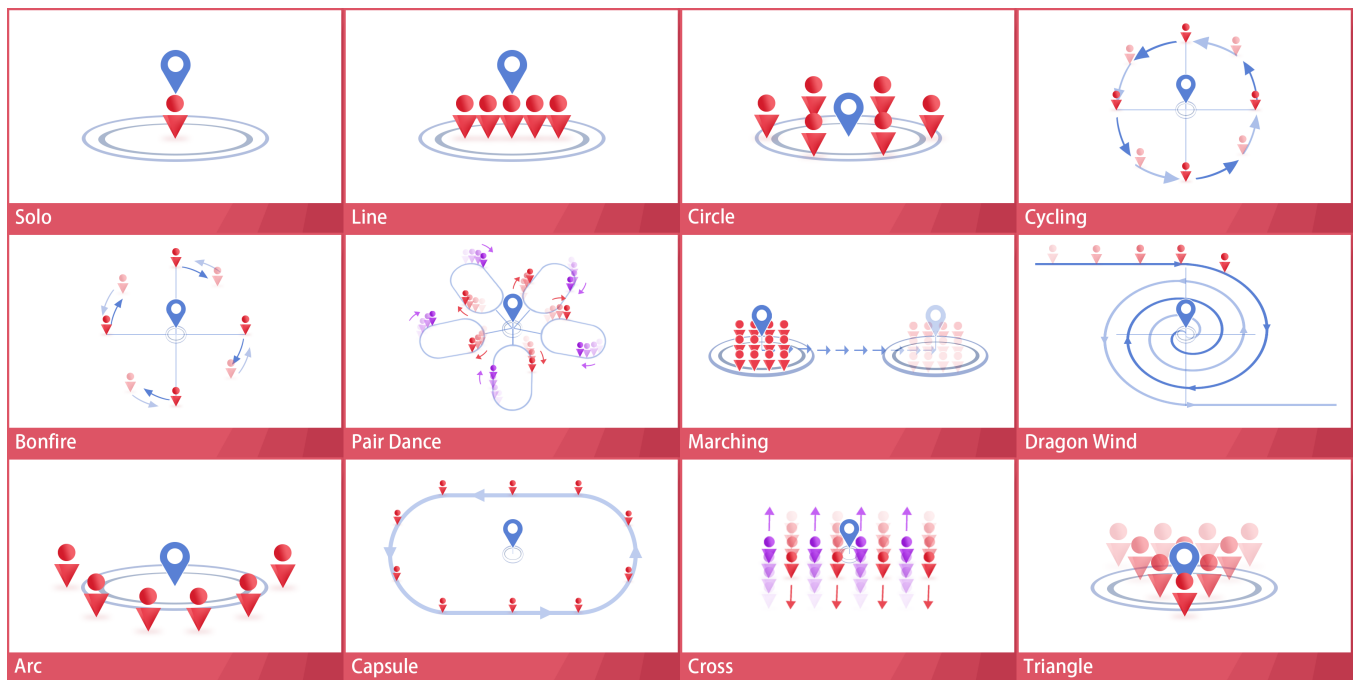


Figure 1: The visualization of 12 elemental dance formations, which are Solo, Line, Circle, Cycling, Bonfire, Pair Dance, Marching, Dragon Wind, Arc, Capsule, Cross and Triangle formation. The blue logo indicates the current position of the formation, and the red logo indicates the position of the dancers in the elemental formation. Best viewed in color.

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ABSTRACT

Intangible Cultural Heritage (ICH) dance refers to traditional dance forms and practices that are recognized as part of a community's ICH, which is a living heritage. However, due to the decrease in the number of dance forms and inheritors, the urgency to protect this particular dance form has become imminent. The formations in dance play a crucial role in conveying the meaning of the dance. In this paper, we research 10 ICH dances in Yunnan Province, summarize their characteristics, and propose a dance formation choreography system by voice, FormationCreator. The dance formations are

expressed parametrically through voice in our FormationCreator, allowing for a more flexible definition and reproduction of these formations. Our research seeks to offer initial insights and fresh design concepts for the field of virtual choreography.

CCS CONCEPTS

• **Human-centered computing**; • **Interaction Design**; • **Virtual Reality**; • **Dance Rehearsal**;

KEYWORDS

Intangible Cultural Heritage Dance, Dance Formation, Virtual Reality, Dance Rehearsal, Information Visualization

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1 INTRODUCTION

Dance choreography and visualization are important in the field of sports exhibitions. There is much research focuses on popular sports like soccer [1, 2, 7, 26], ice hockey [21], table tennis [29], baseball [9, 18] and dance [6, 19].

ICH is an indispensable treasure in human culture, characterized by its rich diversity and expressive depth [8, 10, 22, 25, 30, 32]. However, several challenges make the inheritance and promotion of ICH dance difficult [33–35]. The main challenges are as follows: (1) Most inheritors of ICH are elderly, residing in tribal communities, making it challenging to continuously engage with the public and popularize ICH dance; (2) The emergence of new cultures poses a challenge to the traditional dance, leading to a lack of appreciation among younger generations. (3) ICH dances are predominantly group performances, and the expression of formations requires coordinated efforts from multiple participants.

Several endeavors have attempted to achieve choreography presentation. [6] proposes a choreography system for Latin formation dancing, which focuses on planning, training, and analysis of patterns. It leverages a heatmap way to show the utilization of the dance floor. However, this method ignores that each dancer in the dance formation is an independent individual. [19] proposes the Achi Mukua experience space to create an immersive social space and provide users with a good experience of understanding ICH dance. However, they do not consider the importance of dance formation variations on user experience. Although they build a multi-user communication space, they ignore the significance of dance formations to the essence of dance itself. With the rapidly growing demand for interactive VR and the introduction of MetaVerse [5, 11, 13–16, 20, 27, 28, 31], more and more researchers are trying to use this emerging technology to dance [3, 4, 12, 17, 23, 24].

To mitigate the aforementioned problems, we collect 10 types of ICH dance data, analyze and summarize the dance formations in these dance data, and form 12 dance formations, as shown in Figure 1. At the same time, we build FormationCreator, which can change the formation in real-time through voice directly in the

choreography system, as shown in Figure 2. The effectiveness of our system is then verified using a user study with $n = 40$ subjects.

The main contributions of our system are as follows:

- We collect 10 types of ICH dances from Yunnan Province and analyze the ICH dance formations involved.
- We design an ICH dance choreography system FormationCreator, through which we can easily realize the real-time presentation of ICH dance formations directly via voice commands.
- We conduct a user study to validate the effectiveness of the proposed system for dance formation.

2 FORMATIONCREATOR OVERVIEW AND DESIGN EXPLORATION

In recent years, there have been many works trying to leverage multimedia technologies such as RGB-D cameras and motion capture to create an immersive environment that gives users good experiences. However, most of them do not consider the contributions and creation of dance formations. While using the FormationCreator system, the utilization of voice for formation extraction and design will be discussed in detail in the following sections.

2.1 Deconstruction of the Dance Formation

Since dance has many expression forms, expressing the dance formation in a unified way is of great significance to the design of the dance formation system. That means the various dance formations can be designed by parameters. In this paper, we deconstruct formations into elemental patterns, allowing users to customize their own formations based on these elemental patterns through voice. This approach aims to simultaneously meet the preservation requirements for existing artistic forms in artistic software and allow users to create secondary works based on their personal style.

Through analyzing a large dataset of dance morphological patterns captured by Motion Capture, we abstract and form 12 elemental dance formations (as shown in Figure 1). We deconstruct complex formations and construct elemental formations with good reusability, such as *circular formations*, *marching formations*, etc. Additionally, we design some distinct and unique complex formations, like the *dragon wind formation*. Users have the flexibility to create any number of elemental formations in FormationCreator and arrange them freely.

Most elemental forms are dynamic, with their positions changing over time. Dancers are assigned to these positions to follow the movement of the corresponding points and perform formation changes. In order to control and adjust the number, positions, and movement patterns in each formation, we create specific parameter lists for each type of formation. This allows for diverse user design by fine-tuning parameters for each formation. Users can speak the parameter commands via voice. By systematically adjusting the parameter lists of different formations, users can interweave the movement of dancers across formations in time and space to create more intricate formation changes.

Each elemental formation has a designated start and end time, and users can also choose dance steps to synchronize with music for each elemental formation. In addition to these features, we design three transitional formations, which are static formations

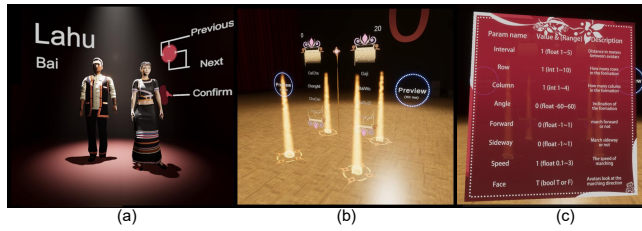


Figure 2: The illustration of the proposed FormationCreator. (a) 10 ICH dance selection interface. It is the dance Bai from Lahu, the two avatars are the dancers from Lahu. (b) is the parameter setting interface for the Marching formation. (c) is the Marching formation parameter list interface. The user can set these parameters through voice. Best viewed in color.

that provide intermediate waypoints for the dancers' movements. Details about these designs are discussed in the next section.

2.2 Dancer Action

In our FormationCreator, users can precisely choreograph the spatial position and movements of each dancer at every moment during the choreography process. To achieve this, we carefully design individual schedules for each enlisted dancer. Once the performance begins, each dancer adheres to the designated schedule to ensure a synchronized and harmonious performance on stage. This method allows meticulous and nuanced control over the unfolding dance narrative, ensuring an engaging and well-coordinated performance for the audience.

Users can set keyframes for each dancer on their independent timeline through formations to control each dancer's position in the performance. Meanwhile, users can select many dancers at once, press the "Assign Place" button, and drag them to a certain point in a particular formation. These dancers will be assigned in sequence to the place chosen by the user. For example, select three dancers and drag them to the second point in the queue direction. Then, the three dancers will be assigned to the second, third, and fourth place in order. If a certain point does not exist, it will not designate a place for the dancer.

When the dancer is assigned to a point, he/she will check the elemental formation associated with that place and record the start and end times of this elemental formation on his/her timeline. By continually assigning dancers to specific points, all dancers will have a timeline recording the start and end times of multiple formations. Once the performance begins, each dancer will act according to the formation start and end times specified on the timeline. When the performance time reaches the start time of a specific formation, the dancers assigned to the current place in that formation will arrive at the designated point. They will then move in accordance with the positions of these formation points until the formation ends. Afterward, the dancers start planning their following path immediately, trying to reach the corresponding point before the subsequent formation begins and repeat the process.

Based on this set of rules, there must be no time overlap between different formations' start and end times. Whenever a dancer is assigned a new position, the dancer autonomously evaluates whether

there is a time conflict with the duration of other formations. In cases of overlap, the action associated with the later start time is automatically discarded to ensure a seamless and non-conflicting choreographic sequence. Users can also use commands to prompt the dancer to "forget" that action.

As mentioned previously, every elemental formation comes with the option to set a specific dance step. When dancers reach their designated positions within the formation, they collectively adopt pre-defined dance steps, creating a coherent and synchronized dance performance. This design choice significantly helps users craft performances with high uniformity and artistic precision. It is worth noting that, to maintain this consistency and choreographic integrity, dancers are not permitted to deviate from the elemental formation during its duration. This restriction ensures that dancers remain seamlessly integrated into the chosen formation, adhering to the specified dance step and contributing to the overall coherence and visual appeal of the performance.

However, the above operations will cause model collision issues. For example, if the dancers change from one elemental formation to another together, they will easily pass through each other (as shown in Figure 3). To solve this problem, we introduce a default waiting formation in the waiting area to ensure a more theatrical and polished presentation. At the beginning of the performance, dancers initiate their movements from designated points within this waiting formation. As the performance concludes, dancers purposefully return to assigned positions within the waiting formation. This design choice adds a layer of completeness to the performance, avoiding the sudden appearance or disappearance of dancers on stage at the very beginning or end of the show. Instead, there is a structured and visually pleasing entrance and exit process. Furthermore, users have the flexibility to influence dancer exits and entrances within the performance. By configuring the timing for dancers to enter or exit the waiting formation, users can effectively control the presence or absence of dancers on the stage. This mechanism alleviates discrepancies in the number of dancers between different formations, allowing users to manage the flow and composition of the performance seamlessly.

2.3 Creation By Voice

We develop a voice control technique to address conflicts between complex system design and sparse VR inputs. Voice commands are similar to what choreographers do on stage in a real-world setting, and they allow real-time modifications. The detailed design of our voice control is shown in Figure 4.

When users raise the controller, they can see a floating indicator in the VR field, denoting the processing stage of voice input. The detailed information of the indicator is as follows: 1. when the indicator is green, it signifies an idle state, allowing the user to input voice commands. 2. when it turns purple, it indicates that the system is recognizing the user's voice commands. 3. when it turns yellow, it denotes that the system is executing operations based on the identified commands. 4. when the system is idle, the user can record their voice by pressing the Grab button on the controller. Upon completing the recording, releasing the Grab button will change the indicator to purple. The system will then recognize the user's voice commands, perform a fuzzy match with the command

Table 1: Instructions list. The instructions for the user to speak to the FormationCreator are given on the left. The right side indicates the corresponding actions taken by the system after receiving the corresponding instructions.

Instruction	Effect
Create dancer	Create a avatar
Change model	Turn selected avatars to another gender
Delete dancer	Make selected avatars off the show
Dismissed	Make selected avatars off the stage
Assemble	Assemble all available avatars
Create [formation name] formation	Create a formation named [formation name] at the given coordinate*
Attach	Assign a position* to selected avatars
Exchange	Exchange selected avatars' route point in selected formation
Detach	Clear the current route point of selected avatars
Forget	Clear all route points of selected avatars
Show timeline	Display the timeline
Next bar	Set time to next bar
Last bar	Set time to Last bar
Time Entry	Set the selected formation to start at the current bar
Time Exit	Set the selected formation end at current bar
Show parameters	Display a list of customizable parameters of selected formation
Add [parameter name]	Set the parameter of the selected formation named [parameter name] plus 0.05 (if is float) or 1 (if is an integer)
Minus [parameter name]	Set the parameter of selected formation named [parameter name] minus 0.05 (if is float) or 1 (if is an integer)
Set [parameter name]	Set the parameter of selected formation named [parameter name] false if is true, true if is false
Play scene	Play the whole scene
Play stop	Stop Play the scene and set the time to 0 bar
Output	Output the play to a video file
Save	Save the scene
Load	Load last saving

table (as shown in Table 1), and execute the command with the highest matching degree. The user can repeat the last command by pressing the bottom button on the controller dial, facilitating repetitive actions such as bulk character creation or specifying multiple positions without the need to articulate the same text repeatedly.

The user's interactions with VR controllers are similar to those in most other software or games. Turning can be achieved by pressing the left or right buttons on the controller dial, teleportation is executed with the upper button, and the user can shoot a laser by pressing the Trigger button to select characters or formations, mark a location, or designate a position in a formation. The system guides the user through relevant operations when commands involving such actions are required.

2.4 Music, TimeLine and Other Setting

We design the "Formation Music" and "Global Music" features to meet the music requirements during the creative process. Each formation, whether elemental or transitional, can be associated with a specific piece of music. The music will start playing when the formation begins and continue until the formation ends. The "Global Music" functionality starts playing at the beginning of the performance and continues until the end. The music for transitional formations is also usable, and due to their flexible start and end times, users can place numerous transitional formations, even those not contributing to the choreography. This innovative approach provides users with a nuanced and powerful tool to choreograph the musical context of their performances, enhancing the overall artistry and immersive quality of choreographed dance presentations.

In order to maintain the fluency and completeness of dance movements, we adopt the "bar" as the fundamental temporal unit within our timeline system. This deliberate choice allows us to fine-tune and synchronize all dance movements to align precisely with whole-number multiples of bars. By adhering to this standardized temporal framework, we aim to eliminate any potential disruptions or incomplete executions of dance actions during the performance, providing a cohesive and visually pleasing choreographic experience for both creators and audiences.

2.5 Save and Export

To enhance user flexibility, we implement a save feature that allows users to pause their work and continue later. The "Save" button generates a local file that is used as the save file for the current scene. When users want to continue the creative process, they can easily select the appropriate save file from the load interface, ensuring seamless continuation of their work. In addition, for users who are satisfied with the works created during choreography after using the FormationCreator, our system provides the function of exporting the dance as a video in recording mode. In this mode, all UI elements are hidden, the user's perspective is temporarily fixed, and the recording contains various perspectives, such as audience perspective, user perspective, and sky perspective. This multi-angle recording method generates multiple video files, allowing users to composite and refine their dance creations for a multi-angle recording effect.

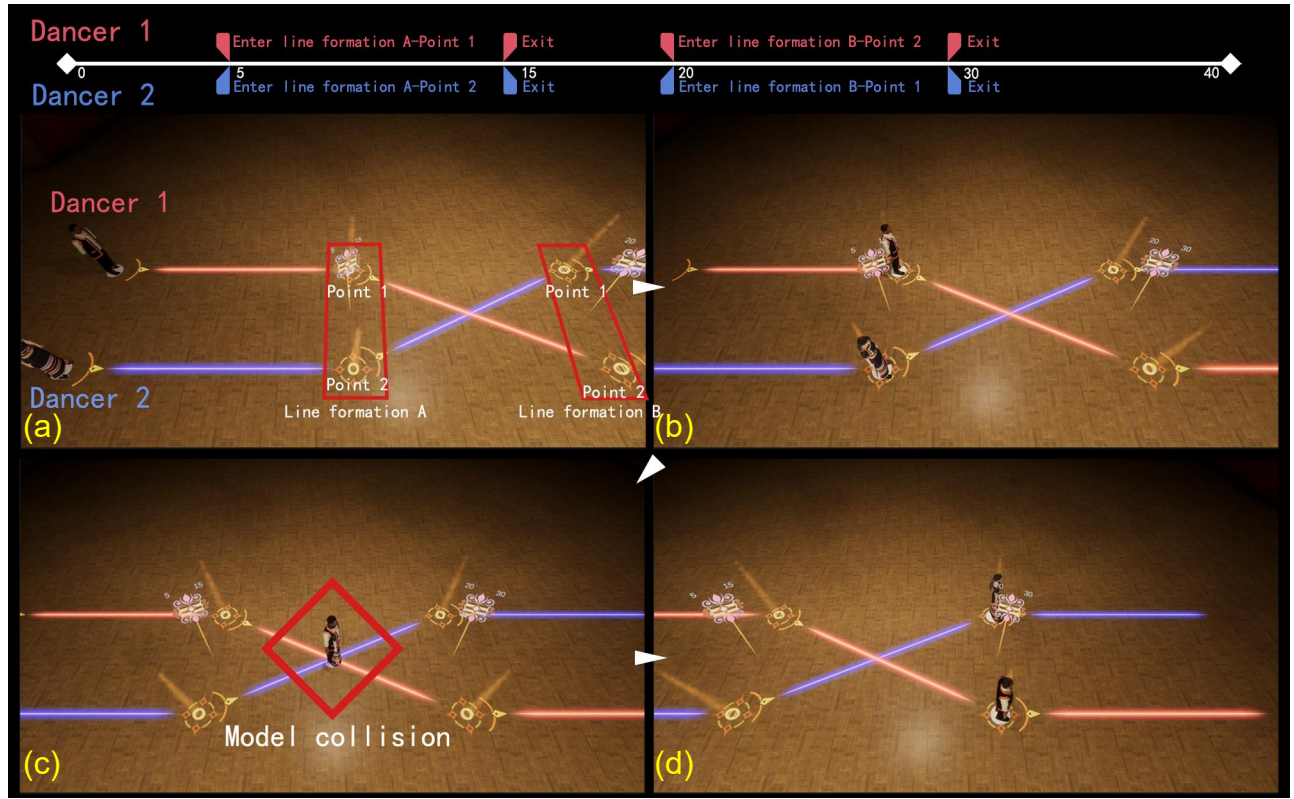


Figure 3: The logic of the dancer's movements. This series of diagrams simulates a duet dance, and the timeline is illustrated at the top. Dancer 1 (shown in light red) and Dancer 2 (shown in light blue) simultaneously enter different positions of the same formation. (a) depicts the initial state of the dance, with two people preparing to enter from the side. Their movement paths are represented by corresponding colored illuminated segments. (b) As the performance progresses, they enter Point 1 and Point 2 of Line Formation A, and subsequently, after the dance concludes, they move into Point 2 and Point 1 of Line Formation B. (c) Due to the interweaving paths, a model collision occurs. In practical choreography, choreographers can employ transitional formations mentioned in our paper to avoid such incidents. (d) shows that Dancer 1 and Dancer 2 enter Point 2 and Point 1 of Line Formation B, respectively. Best viewed in color.

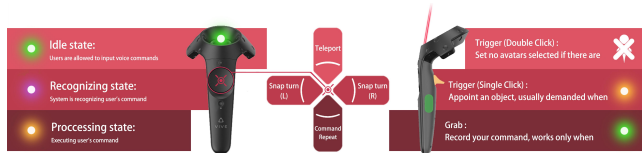


Figure 4: The HTC VIVE controller operations for FormationCreator. We have set corresponding choreography operations for each button on the controller to achieve voice-based ICH dance choreography. Best viewed in color.

3 EVALUATION: USER STUDY

3.1 Research Questions and Measures

This study aims to determine how to enhance our FormationCreator system to make it more useful in demonstrating the beauty of ICH dance. 24 college students (18-22 years old), 10 inheritors from Yunnan Province (55-75 years old), and 6 VR experts (35-50 years old), a total of 40 interviewers (18 of them are female) formed our

interview group. The purpose of the interview group is to improve our system through some simple questions and answers so that more people can understand the ICH dance culture. The following questions will mainly be asked:

- Do you know anything about ICH dance?
- After using this system, have you learned more about the formations of ICH dance?
- Do you think the current system is effective in dance, and are the formations presented enough to attract you?
- Please comment and give some suggestions about our system.

3.2 Result

We have interviewed different groups to evaluate our FormationCreator, and their suggestions give us great inspiration. They all represent this group's views on the construction and display of ICH dance formations by VR systems.

Students from College. Some of the participants are students majoring in digital media, demonstrating a strong interest in gaming, while others are majoring in computer science, showing a keen interest in coding. The combination of these two groups of students provides valuable insights and suggestions for the construction of our system. Almost all students express that the current design and presentation of formations in ICH dances are highly effective and visually appealing. One student points out the novelty and excitement of real-time control over dance formations. They note that in many systems or games they had encountered previously, the control of formations is predefined and lacks the flexibility for spontaneous changes. Another student suggests that not only the dance formations but also the character models and corresponding dance movements within the formations could be changed in real time, providing high flexibility and customization for the users.

ICH Dance Inheritors. The 10 ICH dances are named *Zongshan from Hani*, *Manggu from Hani*, *Dage from Yi*, *Dasanxuan from Yi*, *Tiaocai from Yi*, *Guozhuang from Zang*, *Ligegao from Bai*, *Lusheng from Lahu*, *Bai from Lahu*, and *Archi Mukua from Lisu*. These 10 dances are captured by 10 inheritors using motion capture devices. Most of these inheritors are over 70 years old, and some are even over 80 years old. They do not understand computer technology, so they did not make suggestions for system implementation. However, they are all inheritors of ICH dance. They have a deep understanding of dance. They express unanimous approval for our ICH dance system and believe that the 12 dance formations we gave could cover the 10 ICH dances collected in this article. They even feel that our system offers a more convenient and intuitive way to showcase these dances compared to live performances.

VR Experts. We interview several VR experts to evaluate the effectiveness and demonstrability of our system. Some experts believe that the current system can be a good demonstration of ICH dance formations. However, most of the current dances come from ethnic minorities in Yunnan Province. Each dance actually has its own historical background meaning. It would be more meaningful if some dance introductions could be added to the system. In response to this suggestion, we have added introductions to 10 dances in the system's login interface. Other experts believe that dance formations are a very good form of presentation of dance content. The switching between formations and the communication between dancers within the formations are very important. In response to this suggestion, we generate a transfer formation. Also, since all dance movements in the current version are pre-collected, in subsequent work, we plan to generate richer interactive action content through multi-user access. Another expert suggested that all dance movements and formations in the current system are predefined, and it would be more interesting if they could be customized.

4 LIMITATION AND FUTURE WORK

While our system offers valuable features, particular areas still require attention and improvement. Firstly, the system's coverage of ICH dances is currently limited to 10 types, and future work will be toward diversifying this dataset to encompass a more comprehensive range of ICH dance forms. Secondly, the participant pool primarily consists of students and individuals familiar with ICH, highlighting the need to include multifarious perspectives,

including those unfamiliar with Chinese ICH. Thirdly, the current system for forming dance formations is based on induced patterns from the collected ICH dances, which may limit its generalizability. In future work, we will focus on developing a more dynamic and adaptable system, allowing users to generate dance formations through dynamic descriptions, such as voice or text input. These enhancements aim to address the identified limitations and ensure the system's applicability to a broader user and dance repertoire.

5 CONCLUSIONS

In this research work, we invite students, VR experts and ICH dance inheritors to participate in the design of our system, a VR platform for ICH dance demonstrations. Through the system, common ICH dance formations can be better displayed, allowing users to have an understanding of the ICH dance formations. Through this system, we find that most participating users have a deeper understanding of ICH dance, which plays an essential role in the popularization and dissemination of ICH.

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